

Lab Sheet 4

- Write a python function (with arguments and return statement if required) for the following:
 - max_3(a, b, c)** – returns the max value among a, b and c.
 - even_odd(n)** – prints if the given number is odd or even.
 - Is_prime(n)** – returns True/False accordingly if number is prime or not.
 - Count_prime(n)** – returns the number of prime numbers less than 'n'.
 - Rev_string(s)** – reverses the given string.
 - Rev_num(n)** – reverses a given integer.
 - Fact(n)** – returns the factorial of 'n'.
- Write a Python function that accepts an integer (n) and computes the value of $n+nn+nnn$. (if $n=3$, output is $3+33+333=369$)
- Write a Python function to count the number of vowels in a given string.
- Write a python function to display the type of roots(real or complex) a quadratic equation with coefficients a, b, c will have. (example: **root_type(a, b, c)** = Real roots)
- Write a Python function to compute the greatest common divisor (GCD) of two positive integers.
- Write a Python function to get the least common multiple (LCM) of two positive integers.
- Write a Python function to compute the distance between the points (x1, y1) and (x2, y2) in a plane.
- Write a Python function to calculate a dog's age in dog's years. (Note: For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.) Input: **dog_age(15)** Output: 73
- Write a Python function to check whether a number is perfect or not. *A perfect number is a positive integer that is equal to the sum of its proper positive divisors, that is, the sum of its positive divisors excluding the number itself (also known as its aliquot sum). Equivalently, a perfect number is a number that is half the sum of all of its positive divisors (including itself).* The first perfect number is 6, because 1, 2, and 3 are its proper positive divisors, and $1 + 2 + 3 = 6$. Equivalently, the number 6 is equal to half the sum of all its positive divisors: $(1 + 2 + 3 + 6) / 2 = 6$. The next perfect number is $28 = 1 + 2 + 4 + 7 + 14$. This is followed by the perfect numbers 496 and 8128.
- Write a Python function to find the number of zeros at the end of a factorial of a given positive number. Example: **fact_zeros(11)=2**.

Optional Question:

Write a Python function **strobogram(n)** to print all strobogrammatic numbers that are of length n. *A strobogrammatic number is a number whose numeral is rotationally symmetric, so that it appears the same when rotated 180 degrees.* **strobogram(2)**= 11, 69, 88, 96; **strobogram(3)**= 818, 111, 916, 619, 808, 101, 906, 609, 888, 181, 986, 689